New Generation Data Models and Dbmss Course - Assignment 2 - Crowdfunding Company

Carlo Falchi (Matr. 67340A)[1*]
and Francesco Palladini (Matr. 85652A)[1*]

[1]Department of Computer Science, University of Milan, Italy.

*Corresponding author(s). E-mail(s): carlo.falchi@studenti.unimi.it;
francesco.palladini@studenti.unimi.it;

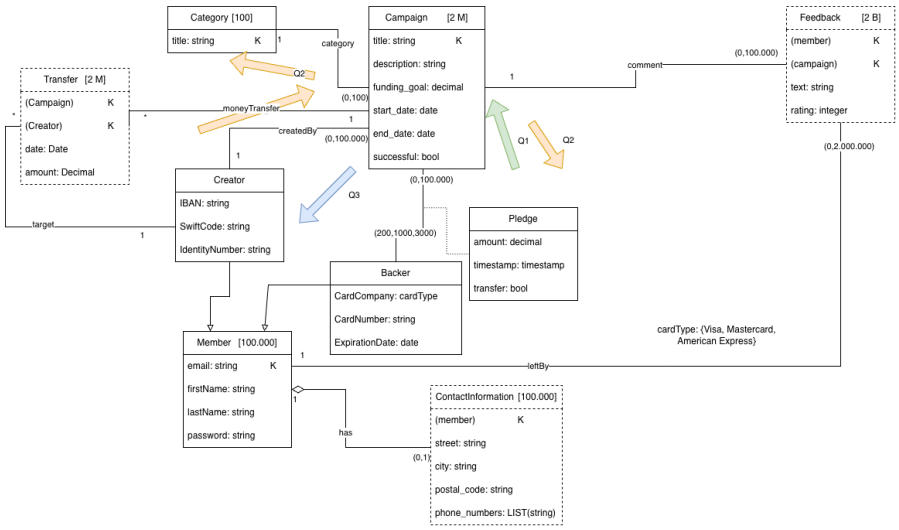


Fig. 1: UML Diagram

1 Solution

1.1 Logical Model

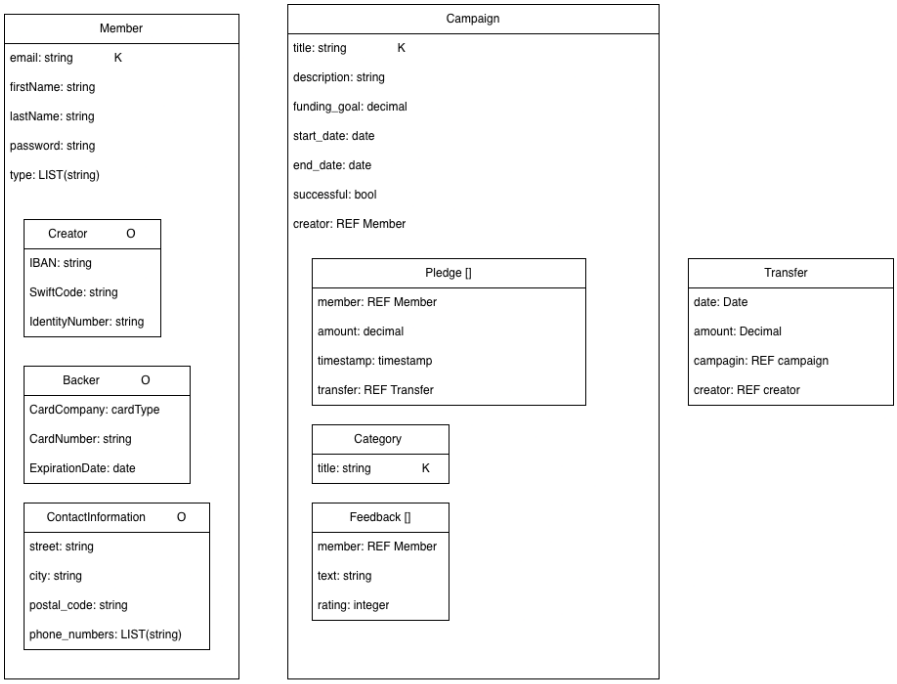


Fig. 2: Logical Model

Workload to consider:

1. Identify the top 5 most competitive campaigns, i.e., those that received the highest number of pledges.
2. For each campaign creator, identify the category in which they achieved the highest average pledge amount.
3. For each month in the last year, identify the top 10 creators per month, i.e., the creators who completed the highest number of successful campaigns in that month

The figure 1 shows the navigational path for the three worloads:

- Q1:(Pledge)→(Campaign)
- Q2:((Creator) $\underset{creator}{\rightarrow}$ (Campaign)→(Pledge) and Campaign $\underset{category}{\rightarrow}$ Category to identify the titles of categories
- Q3:((Campaign) $\underset{createdBy}{\rightarrow}$ (Creator)

The figure 2 reports the local model that we have realized taking into the account the above operations. The first operation uses Pledge as an entry point, the second uses Creator and the third uses Campaign, Q1 and Q2 are in conflict because they have different directions, however the Campaign is used in all workloads. Furthermore given that a Pledge can be assigned to only one Campaign we suggest using Campaign as entry point and thus collection and embedding pledge in it without risk of duplication. To improve the performance of the second operation (Campaign → Category) we embedded category inside the Campaign so we can avoid the lookup operation, it can introduces duplication but it is considered acceptable. Regarding the third operation (Campaign → Creator), which uses Campaign as entry point, it confirms the need to use campaign as a collection and Creator as a reference using a lookup to avoid duplication of Creators. The feedback collection has been embedded within campaign, given that there is a one-to-many relationship between campaign and feedback, and to avoid duplication we use a reference to the Member class. The Transfer collection has been created as a separate collection because it is not included in the most important workloads, we expect that a report or a page will request all transfers rarely so we can insert Campaign and Creator as a reference and use the lookup operation for them. Finally the member collection has been created as a entry point and a separate collection mostly for to make second operation faster, optional contact information has been embedded in it but there are further improvements explained in the next section.

1.2 Design Patterns